Short-answer bank and final pass

Answers and worked solutions · Day 24 · Mon 16 Nov · pack MXF

1. The 20 calculations that come up every year

Q1 territory (Ch1–2): power, AC circuits, three-phase

- 2023 quiz Q1: $P_{OUT} = 1.1$ kW, $P_{IN} = 1.341$ kW, loss 0.241 kW, $\tau = 7.346$ Nm (149.7 rad/s).
- Jun24 Q1b: pf = 0.8184 lag, $P_{OUT} = 3.420$ kW, $\eta = 85.50\%$.

Jun23 Q1a: $E = 13.8$ V, $R_S = 0.01433\ \Omega$, loss 1290 W, motor 2850 W, $V(20\ \text{A}) = 13.51$ V.

Jun23 Q1b (70 Ω, 460 V): star 3.794 A phase and line, 3.023 kW. Delta 6.571 A phase, 11.38 A line, 9.069 kW.

- Jun23 Q1c ($I_1 = 13\angle 20^\circ$ A, 10 Ω, $j30$ Ω, $-j50$ Ω): $V_R = 130\angle 20^\circ$, $V_L = 390\angle 110^\circ$, $V_C = 650\angle -70^\circ$ V. $Z = 22.36\angle -63.43^\circ\ \Omega$, $V_1 = 290.7\angle -43.43^\circ$ V. $S = 1690 - j3380$ VA (sheet: $-j3379$).
- Jun24 Q1c: resistor 400 W and 0 VAr, inductor 0 W and 160 VAr. $V_1 = \sqrt{50^2 + 20^2} = 53.85$ V.
- 2024 test Q3 (parallel, 240 V): $I_R = 2\angle 0^\circ$, $I_L = 4\angle -90^\circ$, $I_1 = 4.472\angle -63.43^\circ$ A. P = 480 W, Q = 960 VAr.
- 2023 quiz Q2c: $V_L = 1.5\angle -20^\circ \times (20 + j5) = 30.93\angle -5.96^\circ$ V.

2023 quiz Q2: $R = 49.31\ \Omega$, $X = 0.9028\ \Omega$. It is inductive (absorbs Q), so $L = 2.874$ mH.

Jun24 Q1d: load pf 0.8 lag. $P_S = 80$ W, $Q_S = 38.75$ VAr, $Q_C = -21.25$ VAr. For $Q_C = -30$ VAr at 115 V, 60 Hz: $C = 6.017\ \mu$F.

Q2 territory (Ch3): magnetics and transformers

- Jun23 Q2a(i): $\mathcal{R}_g = 596.8$ kA/Wb, $\mathcal{R}_c = 66.31$ kA/Wb, $L = 1.544$ mH.
- Jun24 Q2a(i): $\mathcal{R}_g = 810.6$ kA/Wb, $L = 17.77$ mH.
- 2023 quiz Q3b: $\mathcal{R} = 100\,000$ A/Wb, $L = 0.4$ H, $X = 125.7\ \Omega$. Then $I(100\ \text{V}) = 0.7958$ A. The sheet gives 0.7955 A because it rounded X to 125.6 Ω.

- Jun23 Q2a(ii): $E = 3.867$ J.
- Jun24 Q2a(iii): $F = 439.5$ N.
- 2023 quiz Q4: $I = 46.69$ A, $F = 962.9$ N.
- 2024 test Q4: $B = 0.3519$ T (no gap), 0.1086 T (with paper), $B_{min} = 0.0860$ T.

- 2023 quiz Q3a: $N = 360.4$, so 360 or 361 turns.

- 2024 test Q5: $N_1 = 168.9$, so 169 turns. $R = 3.453\ \Omega$, $N_2 = 33.8$, so 34 turns.
- Jun24 Q2a(ii): $V_{max} = 19.62$ V.

Jun23 Q2a(iii): $P_H = 3.84$ W, $P_E = 14.93$ W, total 18.77 W.

Jun24 Q2b (6:1): $Z_1 = 0.3\ \Omega$, $Z_2 = 10.8\ \Omega$, $Z_3 = 25.8\ \Omega$. $I_1 = 0.4651$ A, $I_2 = 2.791$ A, $V_1 = 5.023$ V, $V_2 = 0.8372$ V. These are consistent with a 0.3 Ω load and a 15 Ω series resistance on the primary (see the figure in the PDF).

- Jun23 Q2b: $R = 0.5845\ \Omega$, $X = 2.721\ \Omega$, $R_C = 1280\ \Omega$, $X_M = 186.6\ \Omega$.
- 2024 test Q6: $X = 1.005\ \Omega$ and $R = 0.3068\ \Omega$. **The official sheet prints 0.3086 Ω, which is a digit swap.** $46\,667/390^2 = 0.3068$.

- 2023 quiz Q5: $P_{CU} = 1.179$ kW, $P_{FE} = 1.369$ kW, $\eta = 97.78\%$ (sheet 97.79%, a rounding difference).
- 2023 quiz Q6: (a) $V_2' = 1.905\angle 0^\circ$ kV, $I_2' = 6.056\angle -38.74^\circ$ A. (b) $V_1 = 2.254\angle 5.89^\circ$ kV, so the line voltage is **3.904 kV**. **The sheet's 3.883 kV does not match its own phase value** ($\sqrt{3} \times 2.254 = 3.904$). Read "21000V" as $2100\angle 0^\circ$ V.
- 2024 test Q7: $V_2' = 8.083\angle 0^\circ$ kV, $I_2' = 6.186\angle +13.49^\circ$ kA.
- Jun24 Q2c(ii): $I_1 = 660.9$ A, $P = 1.024$ MW.

Q3 territory (Ch4–5 basics in part a, Ch7 induction machines)

Jun23 Q3a: 10 poles ($n_S = 600$ rpm). Induction, because 560 < 600. Ideal efficiency 93.33%. At ¼ torque: 590 rpm, 26.34% of rated power.

Jun24 Q3b: $R_1 = 2\ \Omega$, $R_2' = 3.2\ \Omega$, $X = 7.635\ \Omega$.

- Jun23 Q3c: 4 poles, 1500 rpm. At 1520 rpm it is a generator. At 1450 rpm, $\tau = 101.7$ Nm and $I_1 = 29.79\angle -38.44^\circ$ A.
- Jun24 Q3c: $n_S = 1000$ rpm, $s = -0.025$, $R_2'/s = -120\ \Omega$. $I_1 = 2.029\angle -174.2^\circ$ A, $P_{IN} = -1451$ W, $P_{AG} = -1.482$ kW, $\tau = -14.15$ Nm.

Jun23 Q3b(i): $R'_{EXT} = 2.828 - 0.5 = 2.328\ \Omega$. With no external resistance, $s_{\tau,max} = 0.177$.

Q4 territory (Ch8 and per-unit)

- Jun23 Q4a: $X_S = 3.676\ \Omega$. $V = 1.0\angle 0^\circ$ pu, $I = 0.7454\angle -26.57^\circ$ pu. The sheet prints 0.7453∠26.57°: it truncates, and the OCR lost the minus sign. Delivering Q means a lagging current.
- Jun24 Q4a: $X_S = 1.047\ \Omega = 2.137$ pu. $V = 0.97\angle 0^\circ$ pu, $I = 0.6443\angle -36.87^\circ$ pu.

> Jun24 Q4b: $I = 1\angle +25.84^\circ$ pu, $E_0 = 1.634$ pu, $\delta = 82.43^\circ$ (sheet 82.42°). $P_{max} = 0.9079$ pu at 90° (sheet 0.9077, from the rounded E_0).

> Jun23 Q4b: induction motor 235.3 kW, 158.1 kVAr. Synchronous motor 222.2 kW, 114.5 kVAr (sheet 114.6, from the rounded 222.2). Plant pf = 0.9955 lag.

2. Short-answer bank (Q4c style)

Past-paper short answers (official points)

S1.

- Many loads, especially induction motors, draw reactive power as well as real power. Reactive power is energy that oscillates back and forth between the source and the load each cycle, and does no useful work.
- It still needs current. So the supply current and the apparent power are larger than the real power alone needs. This can overload generators, transformers and cables, and causes extra resistive losses.
- Power-factor correction connects a capacitor in parallel with the load. The capacitor generates reactive power, so the load's reactive power is supplied locally and the reactive power drawn from the supply falls.
- The real power and the load's own operation are unchanged. Only the supply current falls, and the supply power factor rises towards one. Supply companies also charge for peak volt-amps, so the bill drops.

`OFFICIAL POINTS:` loads draw reactive power, which is energy oscillating in the system that can overload generators and transformers and produce extra losses. A parallel capacitor generates reactive power and reduces the reactive power drawn from the supply.

S2.

- **Similar:** both can run from three-phase AC. Both have a primary winding (the stator), a secondary winding (the rotor) and an iron core. Both transfer power by electromagnetic induction, so their equivalent circuits have the same elements.
- **Different:** the induction machine's secondary rotates and is short-circuited. A transformer's secondary is stationary and feeds an external load.
- The rotor currents are at slip frequency, which changes with speed. In a transformer both windings are at supply frequency.
- In the induction machine equivalent circuit the rotor resistance is divided by the slip. This represents both the rotor copper loss and the mechanical output.
- The induction machine has an airgap, so its magnetising current is a much larger fraction of rated current.

`OFFICIAL POINTS:` both 3ph AC, with primary and secondary windings, a magnetic core and similar equivalent circuits. But the IM secondary is rotating and short-circuited,

and the rotor frequency differs from the stator frequency. In the IM equivalent circuit the rotor resistance is divided by slip.

S3.

- The referred rotor resistance would be zero.
- There would be no rotor copper loss, so the efficiency would rise.
- The slip at maximum torque is proportional to rotor resistance, so it would be zero. The torque–speed curve would be vertical at synchronous speed, like a synchronous machine: no speed drop with load.
- The starting torque would be zero. At standstill the effective rotor resistance absorbs no power, so the motor could not start by itself.

`OFFICIAL POINTS:` zero rotor resistance; the torque–speed slope at synchronous speed becomes vertical; no rotor losses, so higher efficiency; zero starting torque. (Think of the Q3b(ii) curves.)

S4.

- **Real power** is the average of the instantaneous power. It comes from the current component in phase with the voltage. It is the net energy delivered to the load and does useful work (heat, light, shaft power). Resistances absorb it.
- **Reactive power** comes from the current component 90° out of phase with the voltage. It is energy that oscillates between source and load and averages to zero. Inductors absorb it and capacitors generate it. It does no useful work but still needs current.
- **Apparent power** is the magnitude of the complex power, whose real part is the real power and whose imaginary part is the reactive power. It is the rms voltage times the total rms current.
- **Significance:** apparent power sets the current, and so the rating of generators, transformers and cables. Real power is the energy that is paid for. Reactive power is why power-factor correction is used. Real and reactive power are both scalars.

`OFFICIAL POINTS:` for real and reactive power, any of: phase of current relative to voltage, link to instantaneous power, energy flow between source and load, ability to do useful work, which elements are associated, both scalar. For apparent power: magnitude of complex power, related to total current.

S5.

- **Copper loss:** resistive heating of the windings by the load current. It is modelled by the series winding resistance (primary plus referred secondary).
- **Iron loss:** hysteresis loss (energy to cycle the core's magnetisation) and eddy-current loss (currents induced in the core). It is modelled by the core-loss resistance in the shunt branch across the supply.

- **Variation:** copper loss goes with the square of the current, so at constant voltage it goes with the square of the output power. Iron loss depends on the flux, which is set by the supply voltage and frequency, so it stays roughly constant with load.
- So iron loss dominates at light load and copper loss at heavy load. Maximum efficiency occurs where they are equal.

OFFICIAL POINTS: explain copper and iron losses, their physical mechanism and their equivalent-circuit parameters. Copper loss is proportional to the square of output power, and iron loss does not vary with output power.

S6.

- **Magnetising current:** the stator must set up the rotating field even with no load. This current flows in the magnetising reactance. The airgap makes the reluctance high, so this current is large (about 25–60% of rated current, against about 10% in a transformer) and almost purely reactive. That is why the no-load power factor is low.
- **Core loss:** the rotating flux causes hysteresis and eddy-current loss in the iron. A small in-phase current, modelled by the core-loss resistance, supplies this loss.
- **Mechanical loss:** bearing friction and windage (air drag, including the cooling fan) need some shaft power even with no load. So the rotor runs slightly below synchronous speed, and a small rotor current flows.
- These currents also cause a small stator copper loss.

OFFICIAL POINTS: magnetising current and core-loss resistance (their equivalent-circuit parameters and physical reasons), and mechanical losses needing power even at no load, with the types of mechanical loss.

S7.

- **Process:** for a given voltage and frequency, the back-emf equation fixes the product of turns and peak flux. So the number of turns sets the peak flux density. Choose the turns so that the peak flux density is just below the saturation value of the core.
- **Too few turns:** the peak flux density needed is too high, so the core saturates. The magnetising current rises sharply and becomes distorted, the input current is high, and the winding may overheat.
- **Too many turns:** the flux density is lower than needed, so the core is under-used. The longer winding (and thinner wire, to fit the window) has more resistance. Copper loss rises, efficiency falls, and the winding may overheat.

OFFICIAL POINTS: the turns change the peak core flux (back-emf equation), and are chosen so the flux density is below saturation. Too few: too high a flux density, saturation, high input current, possible overheating. Too many: low flux density, higher winding resistance, higher copper loss and lower efficiency, possible overheating.

S8.

- **Ideal:** the output voltage does not change. It is set only by the input voltage and the turns ratio, and the load does not affect it.
- **Ideal:** the output current halves, by Ohm's law, because the voltage is fixed and the resistance has doubled.
- **Real:** the output voltage rises slightly. A real transformer has series winding resistance and leakage reactance. The lower load current gives a smaller voltage drop across them.
- **Real:** the output current still falls, but to slightly more than half, because the output voltage is now slightly higher.

`OFFICIAL POINTS:` the same four points.

S9.

- It **lags**.
- The series impedance of the transformer is mainly leakage reactance, so the load current lags the primary voltage that drives it.
- The load is resistive, so the load voltage is in phase with the load current. It therefore lags the primary voltage.

`OFFICIAL POINTS:` lagging, because the series reactance makes the current lag the primary voltage, and the load voltage is in phase with the current.

S10.

- It is an **induction** machine. 560 rpm is not a synchronous speed at 50 Hz: the nearest one above it is 600 rpm (10 poles).
- A synchronous machine runs exactly at synchronous speed. An induction machine must run slightly below it, so that there is slip and therefore rotor current and torque.

`OFFICIAL POINTS:` induction, as the rated speed is less than the synchronous speed.

S11.

- **DC machine:** halve the armature voltage. The no-load speed is proportional to voltage, and the slope of the torque–speed line stays the same.
- **Synchronous machine:** halve the frequency and halve the voltage. The frequency sets the synchronous speed, and halving the voltage as well keeps the volts-per-hertz ratio, and so the flux, at its rated value.
- **Induction machine:** halve the frequency and halve the voltage, for the same reason. The curve keeps its shape and slides to half speed.

- Never reduce only the frequency on an AC machine: the flux would double, the core would saturate and the current would become very large.

`OFFICIAL POINTS:` synchronous: halve both V and f. DC: halve V. Induction: halve both V and f.

S12.

- On open circuit, only the shunt branch (core-loss resistance in parallel with magnetising reactance) carries current, and it is a fixed impedance in the model.
- Current ×0.5, because current is proportional to voltage.
- Real power ×0.25 (voltage squared over the core-loss resistance). Reactive power ×0.25 (voltage squared over the magnetising reactance).
- Power factor ×1, because both powers scale equally.

`OFFICIAL POINTS:` ×0.5, ×0.25, ×0.25, ×1.

Likely new short answers

S13.

- **Speed:** a synchronous machine runs exactly at synchronous speed at any load. An induction machine runs slightly below it as a motor (above it as a generator), and the slip grows with load.
- **Rotor field:** in a synchronous machine the rotor field comes from DC field current (through slip rings) or permanent magnets. In an induction machine, the rotor currents are induced by the slip between the rotor and the rotating stator field, so no rotor supply is needed.
- **Reactive power:** an induction machine always absorbs reactive power to magnetise its airgap, so it runs at a lagging power factor. A synchronous machine's reactive power is controlled by the field current. It can generate or absorb reactive power.
- **Starting and cost:** an induction motor is self-starting, cheap and robust. A synchronous motor is not self-starting on a fixed-frequency supply and costs several times more.

S14.

- Speed is changed by changing the supply frequency from an inverter, because frequency sets the synchronous speed.
- By the back-emf equation, the peak flux is proportional to voltage divided by frequency (neglecting stator resistance).
- If the frequency is reduced at constant voltage, the flux rises above its rated value and the core saturates. The magnetising current then becomes very large and the machine overheats.

- If the voltage is reduced too much, the flux falls and so does the available torque.
- Keeping the volts per hertz at the rated value keeps the flux constant. So the maximum torque stays about the same, and the torque–speed curve slides along the speed axis. At low speed, the drop across the stator resistance means a small voltage boost is needed.

S15.

- The secondary of both is short-circuited. But in the induction machine the rotor resistance is divided by the slip, so the effective rotor resistance changes with speed.
- This is because the rotor moves. The voltage induced in the rotor, and the rotor frequency, are proportional to the slip. Referring the rotor to the stator frequency turns this into the rotor resistance divided by the slip.
- This resistance represents two things: the real rotor copper loss and the mechanical output power.
- The magnetising reactance is much smaller (the magnetising current much larger), because the flux must cross the airgap. So the shunt branch cannot be ignored the way it often can for a transformer.

S16.

- An airgap has a relative permeability of one, so even a short gap has a reluctance far larger than the iron. The gap usually dominates the total reluctance.
- So the inductance falls, and a larger current is needed for the same flux. Because the gap reluctance is constant, the inductance also becomes more linear and less dependent on the iron's permeability and saturation.
- Most of the magnetic energy is stored in the gap, and the force on the poles comes from the field in the gap.
- In an induction machine the gap is unavoidable, because the rotor must turn. It is made as small as mechanically possible, because a larger gap means more magnetising current, a lower power factor and more stator copper loss.

S17.

- Each quantity is expressed as a fraction of a base value, usually the machine's rating. So you can see at once whether a value is large or small: one per-unit voltage is rated voltage, and one per-unit current is rated current.
- Per-unit values of similar machines fall in a narrow range regardless of size, for example a synchronous reactance of about one to two per-unit. This makes it easy to compare machines and to spot errors.
- In three-phase per-unit, the line and phase values are equal and the total and per-phase powers are equal. The square-root-of-three and factor-of-three terms, and the star–delta conversions, disappear from the working.

- With bases chosen consistently on each side, a transformer's impedances are the same in per-unit whichever side they are referred to. This simplifies power-system calculations.

S18.

- Industrial plants are dominated by induction motors. These must absorb reactive power to magnetise their airgaps, so the plant draws a lagging current.
- Supply companies charge for peak volt-amps as well as energy, so a low power factor costs money. It also loads cables and transformers.
- A synchronous motor's reactive power is set by its field current. If it is over-excited, it generates reactive power (a leading input current) while still driving its mechanical load at constant speed.
- That generated reactive power offsets the induction motors' lagging reactive power, so the total plant power factor rises towards one.
- The amount available is limited by the motor's volt-amp rating less its real power. Synchronous motors cost several times more than induction motors, but the lower tariffs can justify the cost.

S19.

- The machine is connected to the supply and driven by a prime mover above synchronous speed. The slip is then negative.
- The effective rotor resistance becomes negative, so the airgap power and the torque reverse sign. Mechanical power from the shaft is converted to electrical power and fed back to the supply.
- It still needs magnetising current, so it still absorbs reactive power from the supply (or from capacitors) even while it delivers real power. It cannot control its own reactive power or voltage.
- Its output depends on how far above synchronous speed it runs, so it suits wind turbines and regenerative braking. Its losses still have to be supplied, so the electrical output is less than the shaft input.

S20.

- Balanced three-phase stator currents produce a magnetic field of constant magnitude that rotates at synchronous speed.
- When the rotor turns slower than the field, its conductors see a changing flux. This induces voltages in them at slip frequency, and because the rotor is short-circuited, rotor currents flow.
- These currents, in the stator field, produce forces on the rotor conductors, and so a torque in the direction of the field's rotation.

- At synchronous speed the rotor moves with the field and sees no change of flux. There is no induced voltage, no rotor current and no torque. Some slip is always needed.

S21.

- **DC test:** measure the stator winding resistance with DC. This gives the stator resistance per phase.
- **No-load test:** run unloaded at rated voltage. The slip is nearly zero, so the rotor branch draws almost no current, and the shunt branch dominates. Measuring voltage, current and power gives the core-loss resistance and magnetising reactance. The core-loss figure also includes friction and windage.
- **Blocked-rotor test:** hold the rotor still (slip of one) and apply a reduced voltage that gives about rated current. The voltage is low, so the shunt branch can be ignored. The measurements give the total series resistance and leakage reactance.
- Subtract the stator resistance from the total series resistance to get the referred rotor resistance. The leakage reactance is usually split equally between stator and rotor.

S22.

- **Squirrel cage:** bars (usually aluminium) shorted by end rings. Its resistance is fixed. It is cheap, robust and the most common type.
- **Wound rotor:** a three-phase winding brought out through slip rings, so external resistance can be added.
- More rotor resistance increases the slip at which maximum torque occurs, but the maximum torque itself does not change. This raises the starting torque and lowers the starting current.
- The penalty is higher rotor copper loss and lower efficiency while it is in circuit. So the resistance is cut out once the motor is up to speed.

S23.

- At standstill the motor produces a **starting torque**, usually above rated torque, while drawing a large current (several times rated).
- As speed rises, the torque rises to a peak, the **breakdown (pull-out) torque**, typically two to three times rated. It occurs at the slip set by the rotor resistance divided by the stator impedance.
- Between the breakdown point and synchronous speed the curve is almost a straight line, and torque is roughly proportional to slip. This is the stable operating region, and **rated torque** sits in it at a few percent slip.
- The torque falls to zero at synchronous speed and reverses above it (the generating region). If the load exceeds the breakdown torque, the motor slows down and stalls.

S24.

- Three-phase stator currents produce a field that rotates at synchronous speed. The rotor carries its own field, from DC field current or magnets, which is fixed relative to the rotor.
- Steady torque is only produced when the two fields rotate together. So the rotor turns at exactly synchronous speed, locked to the supply frequency.
- The torque depends on the angle between the rotor and stator fields (the rotor angle). As a generator the rotor field leads and drags the stator field. As a motor the stator field leads and drags the rotor.
- Torque rises with the rotor angle up to a maximum at 90°. Beyond that the machine loses synchronism (pull-out), which causes large currents and torque pulsations. Rated load is at about 60° to leave a margin.

S25.

- A low-speed generator, such as a hydro machine, needs many poles to make 50 Hz. Salient (projecting) poles make it easy to fit many poles on a large-diameter rotor, and at low speed the centrifugal stresses are modest.
- A high-speed generator, such as a steam or gas turbine at 3000 rpm, has only two or four poles and very high centrifugal forces. A round, solid rotor with the winding set in slots is mechanically strong and has low windage. Its long, thin shape keeps the diameter, and so the stress, down.

S26.

- The main limit is the stator's volt-amp rating, which is set by stator current heating. The apparent power is the hypotenuse of real and reactive power, so the reactive power available is the square root of rated volt-amps squared minus real power squared.
- So with no real power load (a synchronous compensator), the full rating is available as reactive power. At full real power, no reactive power is available.
- Generating a lot of reactive power needs high excitation, so the rotor field current (rotor heating) also sets a limit.
- Absorbing reactive power needs low excitation. That raises the rotor angle towards 90° and reduces the stability margin, which limits the leading side.

S27.

- The terminal voltage and frequency are fixed by the power system. The real power output is set by the mechanical input from the turbine, not by the field current.
- The field current sets the internal emf. At one particular excitation the output power factor is unity, and the stator current is a minimum for that power.

- **Over-excited:** the generator delivers reactive power and runs at a lagging output power factor.
- **Under-excited:** the generator absorbs reactive power, runs at a leading power factor and has a larger rotor angle, so it is closer to losing synchronism.
- Either way from unity, the apparent power and so the stator current rise. A plot of stator current against field current at constant power is the “V-curve”.

S28.

- Both are a voltage source in series with an impedance, connected to the terminals. The source represents the conversion between electrical and mechanical power.
- In both, the internal emf is proportional to the speed and to the field (excitation) flux, and power flows from the source to the terminals when generating and the other way when motoring.
- The DC machine’s quantities are DC and the series element is the armature resistance. The synchronous machine’s quantities are AC phasors, and the series element is mainly the synchronous reactance, with only a small resistance.
- So in the synchronous machine the phase angle of the emf (the rotor angle) controls real power, and its magnitude controls reactive power. The DC machine has no reactive power.

S29.

- The alternating flux induces voltages in the core itself, which drive eddy currents and cause loss.
- Building the core from thin, insulated laminations breaks up the eddy-current paths and greatly reduces the eddy-current loss. Thinner laminations mean less loss.
- Adding silicon to the steel raises its resistivity, which further reduces eddy currents. It also reduces hysteresis loss.

S30.

- For a given power, the line current is inversely proportional to the transmission voltage.
- The resistive loss in the line is proportional to the square of the current, so raising the voltage greatly reduces the loss. Doubling the voltage quarters the loss.
- The lower current also allows thinner conductors and gives a smaller voltage drop. Transformers make it cheap to step the voltage up for transmission and down again for use.

S31.

- A balanced three-phase winding naturally produces a rotating magnetic field of constant magnitude. AC machines work on this principle, so a three-phase machine is

self-starting. A single-phase machine only produces a pulsating field and needs special starting arrangements.
- The total instantaneous power of a balanced three-phase system is constant. A single-phase machine's power pulsates at twice the supply frequency. So three-phase machines run more smoothly, with less vibration and noise.
- Three-phase also uses the machine material and the supply conductors more efficiently. Single-phase is used only at low power in homes, for simplicity and cost.

S32.

- At standstill the stator field sweeps past the rotor at synchronous speed. The rotor field is first pulled one way and then the other, every half cycle.
- The average torque is zero, and the rotor's inertia stops it following, so the rotor stays still and vibrates.
- So it must be brought up to near synchronous speed by other means before it can lock in. Examples are a pony motor, a starting cage on the rotor (induction start), or an inverter that ramps up the frequency.